FLEET TEAM DIGEST

737MAX-FTD-34-19006

Issue Title : Collins GLU-2100 MMR Software GPS Week 9 Anomaly

Airplane Model	ATA	Minor Model(s)
737MAX	3431-42	-8, -9, -BBJ

Other Model(s)	ECCN
737NG, 777	9E991

Originated Date	Last Revised Date	Created On
11/22/2019	10/22/2020	10/16/2020

Estimated Completion Date	Next Update Date

Revision Description

- Updated Final Action section to provide SB release information.

- Updated Milestones section to provide SB release date.

Description

This FTD provides information regarding the GLU-2100 MMR loss of availability during GPS week 9 (June 9th 2019 to Jun 15th 2019).

Background

At the beginning of the ninth week of the GPS calendar, Boeing received reports from operators regarding the loss of GLU-2100 MMR operational availability, while using GLU-2100 software OPS part number, COL4E-0087-0001 or COL4D-0087-0002. Cycling power to the MMR units did not restore the functionality, and the GPS and ILS functions of the unit were unavailable, resulting in degradation of other various systems which are dependent on the GPS position, such as the Automatic Dependent Surveillance-Broadcast (ADS-B) functionality. The root cause has been identified as a software function that is used to calculate the time offset between GPS time and UTC time that was causing the data output processing thread to lock-up. Collins has indicated an MMR software update is necessary to prevent any future occurrence.

Status

Boeing is working with Collins Aerospace, the manufacturer of the unit, and has determined the root cause of this event was due to a software miscalculation which occurred during the week of June 9th 2019 to Jun 15th 2019, and normal functionality of the unit can be restored after June 16, 2019 @ 00:00 UTC. Condition and recovery of the MMR units will fall into the following three categories:

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- Units that were powered down during GPS week 9 are not affected by this event and will perform normally at next power on.

- Units which were powered on during the transition from GPS week 8 to GPS week 9 (June 9, 2019 @ 00:00 UTC) will need to be sent back to the Collins repair center for NVM reset.

- Units which are in an unknown state, or were powered up during week 9, should refer to Collins Aerospace Service Information Letter (SIL) GLU-2100-19-1 Rev. 1, which provides further details and instructions regarding how to perform an initial evaluation.

Boeing has determined that a combination of the unavailability of all MMR functionality, along with a fault of all communication systems in adverse weather conditions that prevented Visual Flight Rules (VFR), could result in the loss of continued safe flight and landing. Therefore, Boeing has initiated multi-model safety SRP: 777-SRP-34-0165.

A repeat of this event is expected to occur next, during the week of June 11, 2023 0:00 UTC to June 18, 2023 0:00 UTC. However, Boeing is continuing to work with Collins Aerospace to update the GLU-2100 MMR internal software to prevent this and any further occurrences. This software update will resolve the GPS software miscalculation.

An additional improper implementation within this OPS software has been identified as detailed in 737MAX-FTD-34-19012. The Federal Aviation Administration (FAA) has indicated that they will issue regulatory action.

Interim Action

Boeing is updating the Maintenance Planning Data document Section 9: Airworthiness Limitations (AWLs) to include information informing operators of this current software limitation within the GLU-2100.

Final Action

Certification of the updated GLU-2100 MMR OPS internal software, PN: COL4C0087-0003, has been completed and Boeing Alert Service Bulletin, 737-34A3572, has been released to provide retrofit instructions for software installation into the GLU-2100 unit. Accomplishment of the SB is required before June 11, 2023 00:00 UTC to avoid a similar event of GLU-2100 MMR functionality loss however an FAA Airworthiness Directive (AD) may mandate an earlier implementation.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

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Root Cause Established: 14-JUN-2019 (Complete)

SRP Solution Selection: 31-OCT-2019 (Complete)

AWL Publication Updated: 30-OCT-2019 (Complete)

SRP Plan Commit: 20-FEB-2020 (Complete)

Production Certification Complete: 25-MAR-2020 (Complete)

Production Incorporation: L/N 7973

Service Bulletin 737-34A3572 Available: 15-OCT-2020 (Complete)

Supplier Information

Rockwell Collins, Inc. Systems

400 Collins Rd. NE

Cedar Rapids, IA 52498-0001

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Information Letter (SIL)	Collins GLU-2100-19-1	1	06/17/2019
Fleet Team Safety (FTS)	ISS-34-19-29489		
Service Related Problem (SRP)	737MAX-SRP-34-0165		
Fleet Team Digest Article (FTD)	737MAX-FTD-34-19012		
Multi-Operator Message (MOM)	MOM-MOM-19-0653-02B		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
GLU-2100	822-2532-100	Collins Aerospace	

Related Categories

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- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Regulatory : Issues mandated by airworthiness directive or other regulatory guidance material
- SRPs - Safety : Service Related Problems in work by Boeing that have an airplane or personal safety impact

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